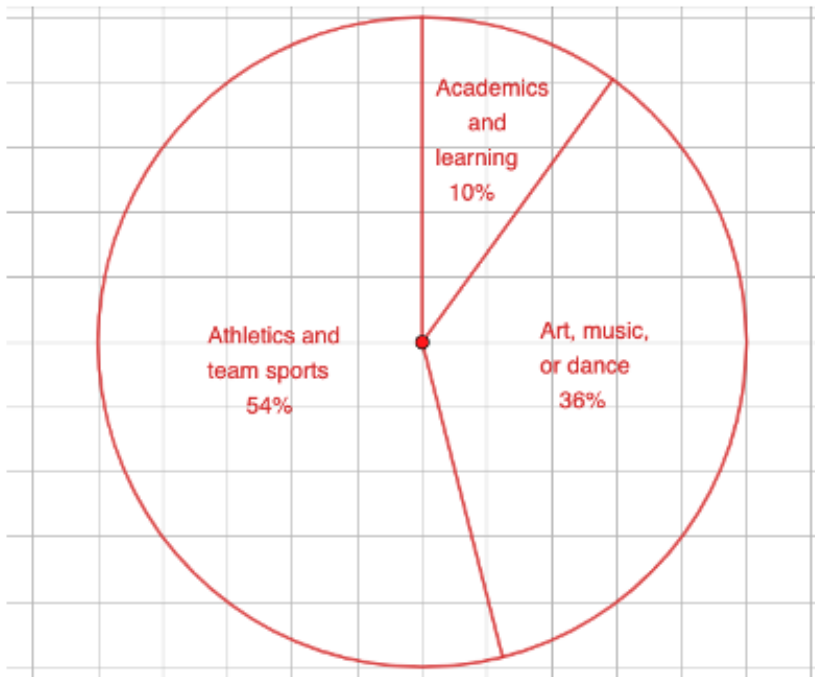


7th Grade Unit 7 Assessment Guide

General Information	Unit 7 uses proportional reasoning, double number lines, and tape diagrams to help build students' understanding of the concepts related to MA.7.AR.3.1 and MA.7.DP.1.4. The two benchmarks in this unit could be difficult for students because of the cognitive demand and the number of steps required. The benchmark MA.7.AR.3.1 covers several different types of questions that involve percentages. An assessment that would cover the variety implied by the benchmark would be too long for a class period. Teachers may wish to use other Math Nation tools such as Check Your Understanding or Test Yourself to further gauge student understanding on specific types of questions. If teachers find that students need support, then Stepping Stones questions may provide students with additional scaffolding. Teachers will find that the students may benefit from occasional review of this standard. As students continue through the 7th grade content, their fluency with rational numbers and proportional relationships should improve, so an intentional spiral review on common errors or common questions may help students strengthen their understanding. MA.7.DP.1.4 requires both proportional reasoning and an understanding of percentages. There are two questions on this assessment for this benchmark. Question 3 uses a survey of 100 students. This question will allow teachers to note if their students are struggling with the basic understanding of the building of a circle graph. Question 4 then asks students to use the given data to build a circle graph. At the beginning of the assessment is a box with the statements: <i>Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.</i> This suggestion can also help a student self-scaffold a problem. As you consider the point distribution, consider the sub-steps students may take. Consider that not every student needs to show sub-steps, so your point distribution should not punish students who can think through the problem and find an answer using a calculator such as the Desmos scientific calculator. It is possible that students can enter these problems as one expression into a calculator such as this.
Calculator Information	The questions on this assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.7.AR.3.1
	\$89.00	Recommended Scoring (10 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 5 points for determining the price paid for one adult and 5 points for the correct answer.
Question	2	Benchmark(s)	MA.7.AR.3.1
	\$8,603.60	Recommended Scoring (10 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 5 points for determining the interest after 8 years and 5 points for the correct answer.

Question	3a	Benchmark(s)	MA.7.DP.1.4
15%		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	3b	Benchmark(s)	MA.7.DP.1.4
112°		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	3c	Benchmark(s)	MA.7.DP.1.4
90°		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	4	Benchmark(s)	MA.7.DP.1.4
$\frac{1323}{2450} = 54\%$ $0.54 \times 360 = 194.4^\circ$ $\frac{882}{2450} = 36\%$ $0.36 \times 360 = 129.6^\circ$ $\frac{245}{2450} = 10\%$ $0.1 \times 360 = 36^\circ$ Students' Preferred Activities (title) 	Recommended Scoring (27 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 3 points for determining the percentage for each category, 3 points for a correct title, 4 points for each correct sector, and 3 points (1 point each) for labeling each sector.	

Question	5	Benchmark(s)	MA.7.AR.3.1
$\frac{282 - 279.2}{279.2} \approx 0.010028 \dots$ $\frac{282 - 285}{285} \approx 0.010526 \dots$ $\frac{282 - 283.06}{283.06} \approx -0.003744 \dots$ Complete the statement. Tool(s) ☐ A only ☐ B only ☐ C only ☐ A and B ☒ A and C ☐ B and C ☐ A, B, and C ☒ do(es) ☐ do(es) not fit within the accepted percent error. Students could also correctly complete the statement with, "Tool(s) B only do(es) not..."		Recommended Scoring (22 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider giving students 6 points for the work for determining whether each tool fits in the error and 4 points for a correct statement.
Question	6	Benchmark(s)	MA.7.AR.3.1
	9,787	Recommended Scoring (10 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.